Test Series No.

05

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

SYLLABUS

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. If h is planck's constant, c is speed of light and λ is wavelength of light, then the dimensional formula of $\frac{\lambda^2}{hc}$ is

(1) $\left[MLT^2\right]$

(2) $\left[M^{-1}LT^2\right]$

(3) $\left[ML^2T^2\right]$

(4) $\left[M^{-1}L^{-1}T^2\right]$

2. A short bar magnet of magnetic moment 0.32 J/T is placed in uniform field of 0.15 T. What is the potential energy of bar magnet in stable equilibrium condition?

(1) -3.2×10^{-2} J

(2) -2.5×10^{-2} J

(3) -4.8×10^{-2} J

(4) -1.5×10^{-2} J

3. If the absolute pressure of a gas enclosed in a container is 8 atm, then its gauge pressure is approximately:

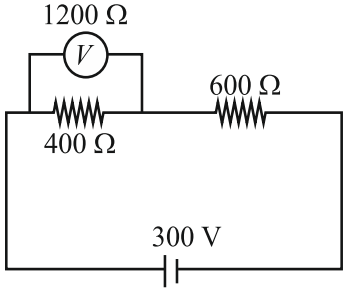
(1) 8×10^5 pascal

(2) 9 atm

(3) 7×10^5 pascal

(4) 6 atm

4. In the circuit shown in figure, the reading of voltmeter of resistance 1200 Ω will be:



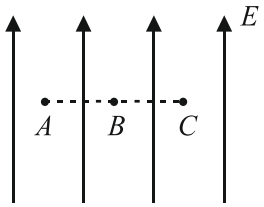
(1) 200 V

(2) 100 V

(3) 50 V

(4) 75 V

5. If A, B and C are three points on a horizontal line in a uniform vertical electric field E, then electric potential is:



(1) maximum at C

(2) same at all the three points A, B and C

(3) maximum at A

(4) maximum at B

6. A constant force acts for 26 s on a body of mass 13 kg on a smooth horizontal surface, starting from rest, after which the force ceases and then body describes 75 m in the next 15 s. The value of force will be;

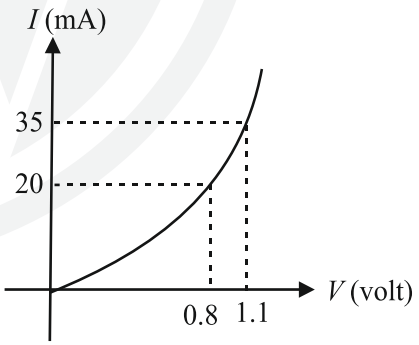
(1) 1.5 N

(2) 2 N

(3) 2.5 N

(4) 3 N

7. The dynamic resistance between 0.8 V – 1.1V for the given semiconductor is:



(1) 20 Ω

(2) 10 Ω

(3) 5 Ω

(4) 0.02 Ω

8. Three stars A, B, C have surface temperatures T_A, T_B, T_C, respectively. Star A appears bluish, star B appears reddish and star C yellowish. Hence

(1) $T_A > T_B > T_C$

(2) $T_B > T_C > T_A$

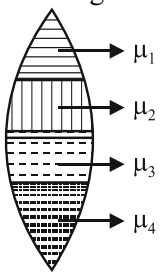
(3) $T_C > T_B > T_A$

(4) $T_A > T_C > T_B$

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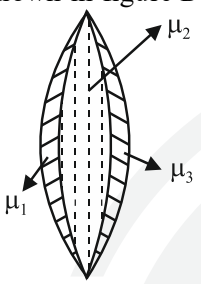
9. Given below are two statements:

Statement I: Number of images of real object formed by a convex lens made of four different material as shown in figure *A* is 4.



(*A*)

Statement II: Number of images of real object formed by a convex lens made of three different materials as shown in figure *B* is 3.

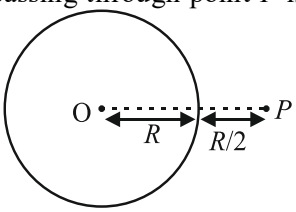


(*B*)

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

10. The radius of gyration of uniform circular disc of mass *M* and radius *R* about an axis perpendicular to disc and passing through point *P* is:



(1) $\sqrt{\frac{11}{4}}R$ (2) $\sqrt{\frac{11}{2}}R$

(3) $\sqrt{\frac{7}{2}}R$ (4) $\sqrt{\frac{7}{4}}R$

11. A conducting circular loop is placed in a uniform magnetic field, $B = 0.025\text{T}$ with its plane perpendicular to the direction of field. The radius of the loop is made to shrink at constant rate of 3 mm s^{-1}. The induced EMF when the radius is 2 cm is:

(1) 2 μV (2) 2π μV

(3) πμV (4) 3π μV

12. The amplitude of a particle executing SHM is made three-fourth keeping its time period constant. If initial total energy of particle is *E*, then its total energy after amplitude is made three-fourth will be:

(1) $\frac{E}{2}$ (2) E

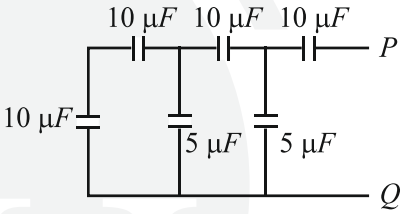
(3) $\frac{3}{4}E$ (4) $\frac{9}{16}E$

13. A drunkard walking in a narrow lane takes 5 steps forward and 3 steps backward, followed again by 5 steps forward and 3 steps backward and so on. Each step is 1 m long and requires 1 s. How much time does the drunkard takes to fall in pit 13 m away from the start?

(1) 49 s (2) 38 s

(3) 50 s (4) 37 s

14. The equivalent capacitance between point *P* and *Q* is:



(1) 10 μF
(2) 20 μF
(3) 5 μF
(4) 15 μF

15. Which of the given statements is/are **correct** for phenomenon of diffraction?

A. For diffraction through a single-slit, the wavelength of wave must be comparable to the size of the slit.
B. The diffraction is very common in sound waves but not so common in light waves.
C. Diffraction is only observed in electromagnetic waves.

(1) A Only
(2) B and C only
(3) A and B only
(4) A, B and C

16. An artificial satellite is revolving around earth in an elliptical orbit. Its:

(1) Kinetic energy is constant
(2) Potential energy is constant
(3) Total energy is constant
(4) Linear momentum is constant

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17. If angle between two vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$ rad, angle between $\vec{a}$ and $-2\vec{b}$ will be:

(1) $\frac{\pi}{3}$ rad (2) $\frac{\pi}{6}$ rad

(3) $\frac{2\pi}{3}$ rad (4) $\frac{5\pi}{6}$ rad

18. The Binding energy per nucleon of $^{7}_{3}Li$ and $^{4}_{2}He$ nuclei are 5.60 MeV and 7.06 MeV, respectively. In the nuclear reaction $^{7}_{3}Li + {}^{1}_{1}H \rightarrow {}^{4}_{2}He + {}^{4}_{2}He + Q$, the value of energy Q released is:

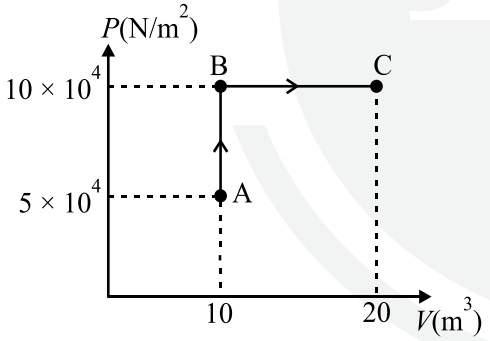
(1) 19.6 MeV

(2) – 2.4 MeV

(3) 8.4 MeV

(4) 17.3 MeV

19. A sample of 2 moles of monoatomic ideal gas is taken through the process ABC. If temperature at A, B and C are T_A, T_B, T_C respectively then $T_A : T_B : T_C$ is:



(1) 1 : 2 : 3

(2) 2 : 3 : 4

(3) 1 : 2 : 4

(4) 4 : 2 : 1

20. Current of 5 A is flowing in a single loop of area 4 m² kept in xz plane. If there is a uniform magnetic field $\vec{B} = \left(2\hat{i} + 2\hat{j} + 4\hat{k}\right)$ tesla, then the magnitude of torque acting on the loop is:

(1) $2\sqrt{320}$ Nm

(2) $4\sqrt{320}$ Nm

(3) $5\sqrt{320}$ Nm

(4) $3\sqrt{320}$ Nm

21. Match **List-I** (type of charge distribution) with **List-II** (variation of electric field with distance r)

List-I		List-II	
(A)	Point charge	(I)	r^0
(B)	Uniformly charged infinite wire (at perpendicular distance)	(II)	$\frac{1}{r}$
(C)	Uniformly charged infinite plane sheet (at perpendicular distance)	(III)	$\frac{1}{r^3}$
(D)	Short electric dipole (at equatorial position)	(IV)	$\frac{1}{r^2}$

(1) A-IV, B-II, C-I, D-III

(2) A-II, B-IV, C-I, D-IV

(3) A-III, B-II, C-I, D-IV

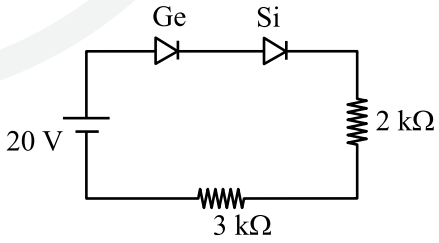
(4) A-III, B-I, C-IV, D-II

22. A standing wave is formed on a string fixed at both ends. The ratio of number of displacement nodes to number of displacement antinodes if string is vibrating in 6th overtone is:

(1) $\frac{8}{7}$ (2) $\frac{7}{6}$

(3) $\frac{9}{8}$ (4) $\frac{2}{3}$

23. In the circuit shown, if the potential drop in forward bias condition across Si and Ge diodes are 0.7V and 0.3V respectively, then the potential difference across 2 kΩ resistor is:



(1) 11.4 V (2) 7.6 V

(3) 8 V (4) 12 V

24. A small particle of mass m is moving towards left with speed 30 m/s. It collides with a very heavy wall which is moving towards right with speed 10 m/s. If collision is perfectly elastic, speed of particle just after collision is:

(1) 60 m/s

(2) 30 m/s

(3) 50 m/s

(4) 100 m/s

[Contd...

25. A transformer having efficiency of 70% is working on 200 V and 5 kW power supply. If the current in the secondary coil is 2 A, then the voltage across the secondary coil is:

(1) 850 V (2) 650 V
(3) 2500 V (4) 1750 V

26. Given below are two statements:

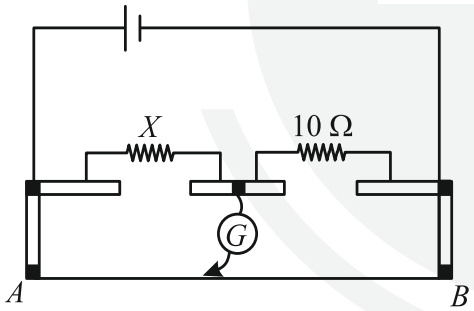
Statement I: Force and plane angle both are dimensional variable.

Statement II : Universal gravitational constant and 'π' both are dimensional constant.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct and Statement II is incorrect.
(4) Statement I is incorrect and Statement II is correct.

27. A meter bridge is set up to determine unknown resistance X using a standard 10 Ω resistor as shown. The galvanometer shows the null point when the tapping key is at a 60 cm mark from point A. Then the value of X is:



(1) 12 Ω (2) 15 Ω
(3) 30 Ω (4) 24 Ω

28. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): In YDSE, the interference pattern disappears, when one of the slits is closed.

Reason (R): Interference occurs due to superposition of light waves from two coherent sources.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

29. Four different bodies A, B, C and D are placed in a room at temperatures 30° C, 90° F, 200 K and 60° C. respectively. Which of these is the coldest body in the room?

(1) A (2) C
(3) B (4) D

30. The maximum length of a uniform wire of density d and breaking stress σ that can be hung vertically such that it does not break under self weight is: (g is acceleration due to gravity)

(1) $\frac{dg}{\sigma}$ (2) $\frac{\sigma g}{d}$
(3) $\frac{\sigma d}{g}$ (4) $\frac{\sigma}{dg}$

31. What value of power factor of series LCR a.c. circuit is **not** possible?

(1) $\frac{1}{3}$ (2) $\frac{1}{4}$
(3) 1 (4) 2

32. Two particles x and y having equal charges and possessing equal kinetic energy enter in a uniform transverse magnetic field and describe circular paths of radius r_1 and r_2 respectively. The ratio of their masses is;

(1) $\left(\frac{r_1}{r_2}\right)$ (2) $\left(\frac{r_1}{r_2}\right)^{\frac{1}{2}}$
(3) $\left(\frac{r_1}{r_2}\right)^{2}$ (4) $\left(\frac{r_2}{r_1}\right)$

33. A rocket of initial mass 8000 kg ejects gases at a constant rate of 16 kg/s with constant relative speed of 10 km/s. The initial thrust on the rocket is:

(1) 1.6×10^5 N (2) 1.6×10^3 N
(3) 3.2×10^5 N (4) 3.2×10^3 N

34. For an equilateral triangular prism, if angle of minimum deviation is δ_m, then refractive index (μ) is equal to:

(1) $\frac{1}{2}\sin\left(30° + \frac{\delta_m}{2}\right)$
(2) $\frac{1}{2}\sin\left(15° + \frac{\delta_m}{2}\right)$
(3) $2\sin\left(30° + \frac{\delta_m}{2}\right)$
(4) $2\sin\left(15° + \frac{\delta_m}{2}\right)$

35. The magnitude of magnetic field strength in a travelling electromagnetic wave in a free space has a peak value of 20×10^{-9} T. The peak value of electric field strength is:
(1) 3 V/m (2) 6 V/m
(3) 9 V/m (4) 12 V/m

36. At 27° C, the rms speed of the molecules of a certain diatomic gas is found to be 1932 m/s. The gas is: (R = 8.3 J/mol-K)
(1) H_2 (2) F_2
(3) Cl_2 (4) O_2

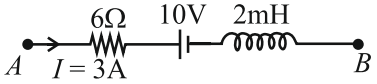
37. If an electron jumps from higher energy level of a H-atom to energy level given in **List-II** then match the **List-I** with **List-II**.

List-I		List-II	
(A)	Balmer series	(I)	$n = 1$
(B)	Paschen series	(II)	$n = 2$
(C)	Brackett series	(III)	$n = 3$
(D)	Lyman series	(IV)	$n = 4$

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-I, B-III, C-IV, D-II
(3) A-II, B-III, C-IV, D-I
(4) A-III, B-IV, C-I, D-II

38. The third overtone produced by a vibrating string (fixed at both ends) 2 m long is 500 Hz. Fundamental frequency of vibration is:
(1) 300 Hz
(2) 125 Hz
(3) 1200 Hz
(4) 200 Hz

39. Network shown in the figure is a part of an electric circuit. At a certain instant, the current I = 3A and it is increasing at the rate of 10^3 A/s. The potential difference between the points A and B ($V_A - V_B$) at this instant is:



(1) 21 V (2) 30 V
(3) 26 V (4) 7 V

40. Choose the option containing **correct** statements.
A. If magnitude of average velocity and average speed over a time interval are same, then particle must move with zero acceleration.
B. Equation of kinematics (motion) can be used for object having non-uniform acceleration.
C. If the displacement covered by a particle is zero, then its distance covered may or may not be zero.
(1) A and B only (2) A, B and C
(3) C only (4) A and C only

41. Two charges of magnitude +2 μC and –50μC are placed 80 cm apart. Calculate the distance of the point from the smaller charge where the electric field intensity is zero.
(1) 20 cm (2) 30 cm
(3) 100 cm (4) 5 cm

42. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: According to Bohr's atomic model, the ratio of angular momenta of an electron in first excited state to electron in ground state is 2 : 1.
Reason R: In a Bohr's model, the angular momentum of the electron is inversely proportional to the principal quantum number.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

43. A convexo-concave divergent lens is made of glass of refractive index 1.5 and focal length 36 cm. Radius of curvature for one surface is three times that of the other. Radii of curvature for the two surfaces are (in cm):
(1) 6, 18 (2) 12, 36
(3) 3, 9 (4) 18, 54

44. A tank has an orifice near its bottom. The volume of the liquid flowing per second out of the orifice does not depend upon:
(1) Area of the orifice
(2) Height of the liquid column above the orifice
(3) Density of liquid
(4) Acceleration due to gravity

45. A particle moves with velocity $\vec{v} = \left(2\hat{i} - 3\hat{j}\right)$ m/s under the influence of a constant force $\vec{F} = \left(2\hat{i} + \hat{j} - 3\hat{k}\right)$N. The instantaneous power delivered to the particle is;
(1) 1 watt
(2) 5 watt
(3) 3 watt
(4) 2 watt

46. Which mixture shows positive deviation from Raoult's law?
(1) Hexane + heptane
(2) Ethanol + water
(3) Acetone + chloroform
(4) Benzene + toluene

47. The number of Faradays (F) required to produce 13.5 g of Al from molten $AlCl_3$ (Atomic mass of Al = 27) is:
(1) 0.5 F (2) 1.0 F
(3) 1.5 F (4) 2.0 F

48. For a reaction, the slope of the Arrhenius plot (ln k vs $\frac{1}{T}$) is -9.0×10^3 K. Find E_a. (R = 8.314 J $mol^{-1}K^{-1}$)
(1) 74.8 kJ mol^{-1}
(2) 9.0 kJ mol^{-1}
(3) 41.5 kJ mol^{-1}
(4) 83.1 kJ mol^{-1}

49. The formula of pentaamminechloridocobalt(III) chloride is:
(1) $[Co(NH_3)_5Cl]Cl_3$
(2) $[Co(NH_3)_5Cl]Cl_2$
(3) $[Co(NH_3)_5Cl]Cl$
(4) $[Co(NH_3)Cl_5]Cl_2$

50. Which of the following amines will give a product soluble in aqueous NaOH on treatment with benzenesulphonyl chloride?
(1) Trimethylamine
(2) Diethylamine
(3) Ethylamine
(4) Ethylmethylamine

51. Match **List-I** with **List-II**.

List-I (Vitamin)		List-II (Deficiency disease)	
A.	Vitamin A	I.	Scurvy
B.	Vitamin B_1	II.	Beri-beri
C.	Vitamin C	III.	Xerophthalmia
D.	Vitamin D	IV.	Rickets

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-II, C-III, D-I

52. If the molality of a solution is increased from 0.20 m to 0.60 m, the value of K_f will be:
(1) tripled
(2) halved
(3) increased by 0.40
(4) unchanged

53. The specific conductance (conductivity) of a 0.2 N NaCl solution at 25°C is 0.015 S cm^{-1}.
The resistance of the cell containing this solution at the same temperature is 40Ω. The cell constant is:
(1) 0.25 cm^{-1} (2) 0.60 cm^{-1}
(3) 1.50 cm^{-1} (4) 2.67 cm^{-1}

54. Given below are two statements:
Statement I: PH_3 is more basic than NH_3 because P is less electronegative.
Statement II: PH_3 has a smaller bond angle than NH_3.
In the light of the above statements, choose the correct answer:
(1) Statement I is false but Statement II is true
(2) Statement I is true but Statement II is false
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

55. How many molecules are in 4 g of H_2? (Molar mass = 2 g/mol, Avogadro number = 6.022×10^{23})
(1) 6.022×10^{23}
(2) 9.033×10^{23}
(3) 1.2044×10^{24}
(4) 3.011×10^{23}

56. Calculate the boiling point elevation of a solution containing 6 g of urea (NH_2CONH_2) in 100 g of water. (K_b = 0.52 K kg/mol , Molar mass of urea = 60 g/mol)
(1) 0.26 K
(2) 0.52 K
(3) 1.04 K
(4) 0.13 K

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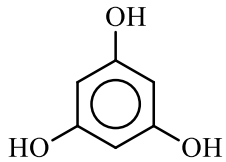
57. For the cell reaction:
$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
$E°_{cell} = 1.10$ V at 298 K. Calculate $\Delta G°$.
($F = 96500$ C mol^{-1})
(1) –106.15 kJ mol^{-1}
(2) +106.15 kJ mol^{-1}
(3) –53.08 kJ mol^{-1}
(4) –212.30 kJ mol^{-1}

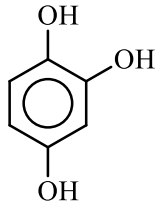
58. In the following sequence of reactions,

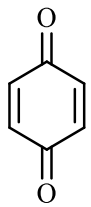
$$CH_3–Br \xrightarrow{KCN} A \xrightarrow[(excess)]{H_3O^+} B \xrightarrow[ether]{LiAlH_4} C$$ the

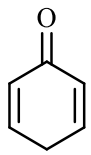
end product C is:
(1) acetone
(2) methane
(3) acetaldehyde
(4) ethyl alcohol

59. For $FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$, $K_p = 0.5$ at 1000 K. If initial $p_{CO} = 1$ atm, what is p_{CO_2} at equilibrium?
(1) 0.5 atm
(2) 0.33 atm
(3) 1.0 atm
(4) 0.67 atm

60. Oxidation of phenol with chromic acid produces:
(1) 

(2) 

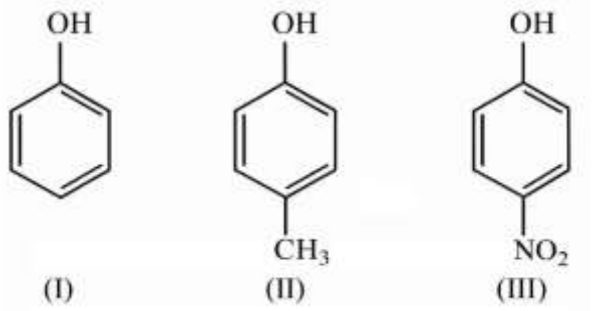
(3) 

(4) 

61. Which of the following is a basic amino acid?
(1) Glycine
(2) Aspartic acid
(3) Arginine
(4) Phenylalanine

62. Which of the following is a combination redox reaction?
(1) $C + O_2 \rightarrow CO_2$
(2) $2HgO \rightarrow 2Hg + O_2$
(3) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
(4) $2H_2O_2 \rightarrow 2H_2O + O_2$

63. A salt solution gives a white precipitate with dilute HCl that dissolves on passing excess H_2S gas, forming a yellow precipitate. The anion present is most likely:
(1) Cl^-
(2) SO_4^{2-}
(3) S^{2-}
(4) $S_2O_3^{2-}$

64. The **correct** acidic order of following is:


(1) I > II > III
(2) III > I > II
(3) II > III > I
(4) I > III > II

65. Which electrolyte has the same van't Hoff factor i as $Al_2(SO_4)_3$ (assuming 100% ionisation)?
(1) Na_2SO_4
(2) K_3PO_4
(3) $CaCl_2$
(4) $K_4[Fe(CN)_6]$

[Contd...

66. Match **List-I** with **List-II**.

List-I (Oxide)		List-II (Nature)	
(A)	NO	(I)	Basic
(B)	SO_3	(II)	Neutral
(C)	Bi_2O_3	(III)	Amphoteric
(D)	As_2O_3	(IV)	Acidic

Choose the **correct** answer from the options given below:

(1) A-IV; B-I; C-II; D-III
(2) A-II; B-I; C-IV; D-III
(3) A-I; B-III; C-IV; D-II
(4) A-II; B-IV; C-I; D-III

67. For the reaction $A+B \rightarrow$ Products, the rate becomes 4 times when $[A]$ is doubled (keeping $[B]$ constant). The rate becomes 2 times when $[B]$ is doubled (keeping $[A]$ constant). The rate law is:

(1) $r = k[A][B]$ (2) $r = k[A]^2[B]$

(3) $r = k[A][B]^2$ (4) $r = k[A]^2[B]^2$

68. The number of geometrical isomers for $[Pt(NH_3)_2Cl_2]$ is

(1) 3 (2) 4
(3) 1 (4) 2

69. Given below are two statements:

Statement I: All Group 16 elements except oxygen form dichlorides and dibromides.

Statement II: All halogens are colourless.

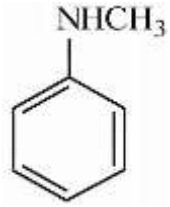
In the light of the above statements, choose the correct answer:

(1) Statement I is false but Statement II is true
(2) Statement I is true but Statement II is false
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

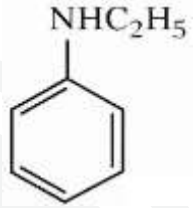
70. Which one of the following sets of monosaccharides forms lactose?

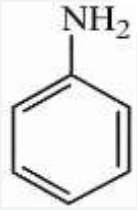
(1) α-D-glucopyranose and β-D-fructofuranose
(2) β-D-galactopyranose and β-D-glucopyranose
(3) α-D-galactofuranose and α-D-glucopyranose
(4) β-D-glucopyranose and α-D-fructopyranose

71. Which of the following amine will give the carbylamine test?

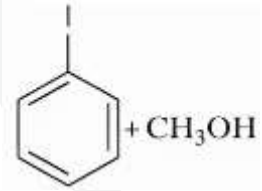
(1) 

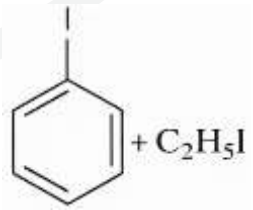
(2) 

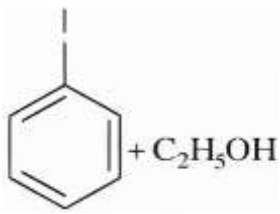
(3) 

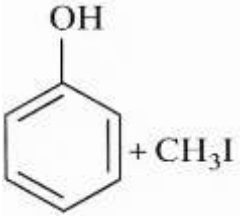
(4) 

72. Anisole on cleavage with HI gives:

(1) 

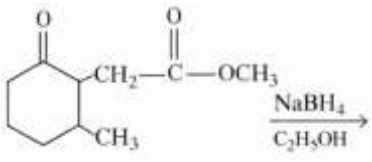
(2) 

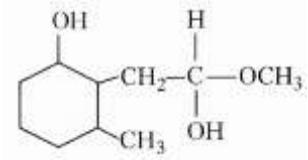
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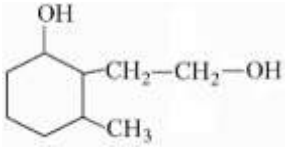
(4) 

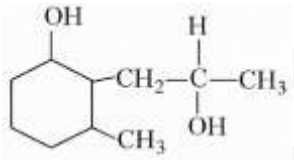
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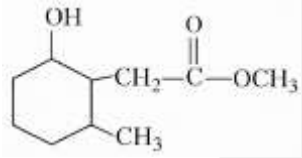
73. The product formed in the following chemical reaction is



(1) 

(2) 

(3) 

(4) 

74. What is the IUPAC name of the organic compound formed in the following chemical reaction?

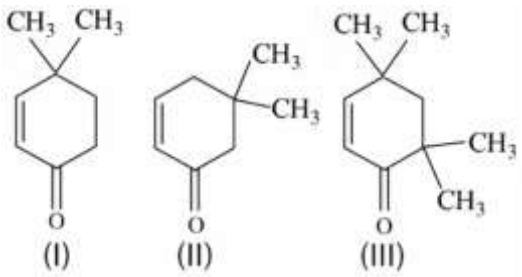
Acetone $\xrightarrow[\text{(ii) } H_2O, H^+]{\text{(i) } C_2H_5MgBr, \text{Dry ether}}$ product.

(1) 2-methyl propan-2-ol
(2) pentan-2-ol
(3) pentan-3-ol
(4) 2-methyl butan-2-ol

75. Which aqueous solution will have the minimum freezing point? (Assume complete ionisation where applicable)

(1) 0.010 m NaCl (2) 0.006 m $MgCl_2$
(3) 0.012 m glucose (4) 0.004 m Na_2SO_4

76. Given,



Which of the given compounds can exhibit tautomerism?

(1) I and II (2) I and III
(3) II and III (4) I, II and III

77. What is the bond order of C_2 according to molecular orbital theory?

(1) 1 (2) 3
(3) 1.5 (4) 2

78. Gadolinium belongs to 4f series. It's atomic number is 64. Which of the following is the **correct** electronic configuration of gadolinium?

(1) [Xe] $4f^8\ 6d^2$
(2) [Xe] $4f^9\ 5s^1$
(3) [Xe] $4f^7\ 5d^1\ 6s^2$
(4) [Xe] $4f^5\ 6d^2\ 6s^2$

79. An organic compound A (C_4H_9Cl) on reaction with Na/diethyl ether gives a hydrocarbon which on monochlorination gives three monochloro derivative (excluding stereoisomers), then A is:

(1) *tert*-butyl chloride
(2) *sec*-butyl chloride
(3) *iso*-butyl chloride
(4) *n*-butyl chloride

80. What is the sign of work done when a gas is compressed at constant pressure?

(1) Negative
(2) Positive
(3) Zero
(4) Depends on temperature

81. Given below are two statements:

Statement I: Complete hydrolysis of XeF_6 gives XeO_3 and HF.

Statement II: Noble gases have low boiling points because strong hydrogen bonding is present.

In the light of the above statements, choose the correct answer:

(1) Statement I is false but Statement II is true
(2) Statement I is true but Statement II is false
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

82. Match **List-I** with **List-II**.

List-I		List-II	
A.	4-Methylphenol	I.	Resorcinol
B.	Benzene-1,2-diol	II.	p-Cresol
C.	Benzene-1,3-diol	III.	Catechol
D.	Benzene-1,4-diol	IV.	Hydroquinone

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-I, D-II

83. Which alkyl halide gives the best yield of amine in Gabriel synthesis?

(1) tert-butyl bromide
(2) chlorobenzene
(3) methyl iodide
(4) vinyl chloride

84. The K_a of HF is 6.8×10^{-4}. What is the pH of a 0.1 M HF solution? (log 8.25 = 0.916)

(1) 1.5 (2) 3.0
(3) 2.5 (4) 2.1

85. For a first-order reaction, $k = 2.31 \times 10^{-3}\ s^{-1}$. Time required for concentration to become one-eighth of initial value is:

(1) 600 s (2) 900 s
(3) 1200 s (4) 300 s

86. Among the following complexes, the one which shows zero crystal field stabilisation energy (CFSE) is

(1) $[Mn(H_2O)_6]^{3+}$ (2) $[Fe(H_2O)_6]^{3+}$
(3) $[Co(H_2O)_6]^{2+}$ (4) $[Co(H_2O)_6]^{3+}$

87. Which of the following undergoes nucleophilic substitution fastest by S_N1 mechanism?

(1) Benzyl chloride (2) Ethyl chloride
(3) Chlorobenzene (4) Isopropyl chloride

88. Given below are two statements: one is labelled as Assertion (A) and the other as Reason (R).

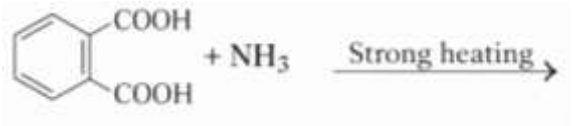
Assertion (A): When bromoethane reacts with alcoholic KCN, the major product is propionitrile (C_2H_5CN)

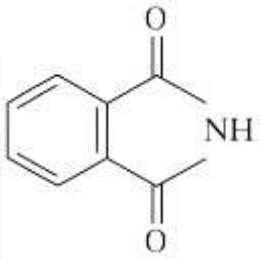
Reason (R): In KCN, the cyanide ion is predominantly ionic and attacks mainly through carbon to form a C–C bond.

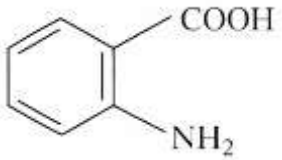
In the light of the above statements, choose the **correct** answer from the options given below:

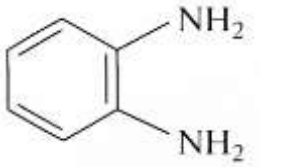
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

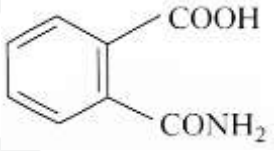
89. The major product of the following reaction is:



(1) 

(2) 

(3) 

(4) 

90. Match **List-I** with **List-II.**

List-I (Aquated ions)		List-II (Number of unpaired electrons)	
A.	Ti^{3+}	I.	0
B.	V^{2+}	II.	1
C.	Mn^{2+}	III.	3
D.	Zn^{2+}	IV.	5

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-I, D-II

[Contd...

91. After agarose gel electrophoresis, separated DNA bands can be visualized by:
(1) Staining with ethidium bromide and exposure to visible light.
(2) Staining with ethidium bromide and exposure to UV radiation.
(3) Direct observation under a light microscope without staining.
(4) Staining with methylene blue and exposure to infrared radiation.

92. Identify the site of glycolytic pathway in case of human and yeast.
(1) Cytoplasm in yeast whereas mitochondria in humans.
(2) Mitochondria in yeast whereas cytoplasm in humans.
(3) Mitochondria in both yeast and humans.
(4) Cytoplasm in both yeast and humans.

93. Read the following statements and identify the **correct** statement.
(1) Steroidal hormones are synthesised in smooth endoplasmic reticulum.
(2) Rough endoplasmic reticulum performs the function of packaging of material.
(3) In plants the contractile vacuole is important for osmoregulation and excretion.
(4) Perinuclear space contains nucleolus and chromatin.

94. How many of the following statements is/are **incorrect**?
A. Sodium-potassium pumps are example of passive transport.
B. The quasi-fluid nature of lipid enables lateral movement of proteins within lipid bilayer.
C. In human beings, the membrane of the erythrocyte has approximately 40 per cent protein and 52 per cent lipids.
D. Both facilitated diffusion and active transport are energy dependent processes.
E. Neutral solutes may move across the membrane by simple diffusion.
Choose the *most appropriate* answer from the options given below:
(1) One only
(2) Three only
(3) Four only
(4) Two only

95. Which of the following statements about adaptive radiation is **correct**?
(1) Adaptive radiation occurs when a single species rapidly diversifies into multiple forms to occupy different ecological niches.
(2) Adaptive radiation is only observed in finches and marsupials.
(3) Adaptive radiation requires the presence of a geographical barrier to occur.
(4) Adaptive radiation results in the evolution of identical species in different geographical areas.

96. Match **List-I** with **List-II** regarding polymerase chain reaction.

List-I		List-II	
(A)	Denaturation	(I)	Used in extension of primers
(B)	Annealing	(II)	Separation of DNA strands by heating at ~94°C
(C)	Primers	(III)	Chemically synthesized oligonucleotides complementary to DNA regions
(D)	Thermostable DNA polymerase	(IV)	Attachment of primers to complementary sequences at template DNA

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-IV, C-III, D-I
(4) A-III, B-II, C-IV, D-I

97. RuBisCO enzyme is:
(1) found only in mesophyll cell of C_4 plants.
(2) present in both C_3 and C_4 plants.
(3) having greater affinity for oxygen when CO_2 concentration is more than O_2.
(4) found in bundle sheath and mesophyll cell of C_3 plants.

98. In a field experiment, when all *Pisaster* starfish were removed from an enclosed intertidal area, the result was:
(1) increase in diversity of invertebrates.
(2) extinction of many invertebrate species.
(3) inability of the *Pisaster* to enter the area again.
(4) replacement of *Pisaster* by other starfish.

99. Find the **mismatched** pair from the following.
(1) Prion - Consists of abnormally folded protein
(2) Chrysophytes - Eukaryotic
(3) Ascomycetes - Coenocytic
(4) Viroids - Smaller than viruses

100. Read the following and select the **correct** option.
(1) In terrestrial ecosystem smaller fraction of energy flow occurs through detritus food chain (DFC).
(2) Interconnection of food webs makes food chain.
(3) DFC is not at all connected with grazing food chain at any level.
(4) Decomposers are mainly fungi and bacteria.

101. Which of the following are **not** characteristic features of sponges?
A. Cellular level of organization.
B. Presence of a water transport system with ostia, spongocoel, and osculum.
C. Direct development without any larval stage.
D. Digestion is extracellular.
E. Reproduction via sexual mode only.
(1) A, B and C
(2) B and D only
(3) C, D and E
(4) C and D only

102. Choose the **correctly** matched pair from the options given below.
(1) Exocrine glands → Secrete hormones
(2) Endocrine glands → Release mucus, saliva, and digestive enzymes through ducts.
(3) Cuboidal epithelium → Found in the lining of the stomach and intestine
(4) Glandular epithelium → Includes goblet cells

103. Given below are two statements:
Statement I: Bacteriophages have a high copy number of their genome within bacterial cells.
Statement II: Bacteriophages and plasmids can replicate independently of chromosomal DNA, but only plasmids can be used as vectors for cloning.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

104. Match **List-I** with **List-II**.

List-I		List-II	
(A)	XO male	(I)	Honey bee
(B)	XY male	(II)	Bird
(C)	Haploid male	(III)	Grasshopper
(D)	ZZ male	(IV)	*Drosophila*

Choose the **correct** answer from the options given below.
(1) A-II, B-IV, C-I, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-II, C-I, D-III

105. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Insulin cannot be effectively administered orally to diabetic patients.
Reason R: Insulin is a protein hormone that gets degraded by digestive enzymes in the gastrointestinal tract.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

106. Which of the following is **true** regarding the *vasa recta*?
(1) It is a network of capillaries formed by the afferent arteriole.
(2) It is a U-shaped capillary network that runs parallel to the loop of Henle in cortical nephrons.
(3) It is responsible for filtering blood in the glomerulus.
(4) It is a U-shaped capillary network that runs parallel to the loop of Henle in juxtamedullary nephrons.

107. A histology student observes a tissue sample under a microscope. It shows fibroblasts scattered among randomly arranged collagen fibers. What is the most likely tissue type?
(1) Dense regular connective tissue
(2) Dense irregular connective tissue
(3) Areolar connective tissue
(4) Adipose tissue

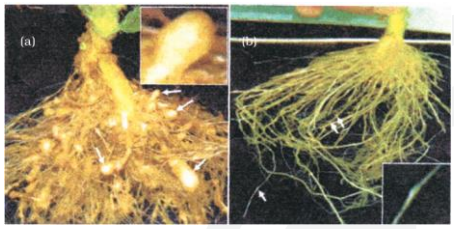
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108. Which of the following statements about the structure of a sarcomere are **correct**?

A. The I-band contains only actin filaments.
B. The A-band contains both actin and myosin filaments and is bisected by the M-line.
C. The H-zone is the region of the A-band where only myosin filaments are present.
D. The Z-line is located at the center of the A-band and anchors the thick filaments.

(1) A only
(2) B and D only
(3) A and C only
(4) A, B and C

109. Carefully observe the diagram given below.



Which of the following **best** explains why the transgenic plant in (b) exhibits a healthier root system despite nematode exposure?

(1) The transgenic plant lacks the receptors needed for nematode recognition
(2) The transgenic plant produces a toxin that kills nematodes upon contact
(3) The transgenic plant expresses dsRNA that silences essential nematode genes, preventing infestation
(4) The transgenic plant has a modified root structure that physically prevents nematode penetration

110. Read the following statements about Hardy-Weinberg equilibrium.

A. The sum total of all allelic frequencies in a population always remains equal to 1.
B. The Hardy-Weinberg equation is represented as $p^2 + pq + q^2 = 1$, where p and q are allele frequencies.
C. Hardy-Weinberg equilibrium applies to all populations, regardless of factors like gene flow and natural selection.
D. A deviation from Hardy-Weinberg equilibrium suggests that evolutionary forces are at work.

Select the **incorrect** statements:

(1) A and D
(2) B and C only
(3) C and D
(4) A, B, and C

111. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Purkinje fibres	(I)	Conducts electrical impulses from the AV node to the ventricles
(B)	Pericardium	(II)	Anchor the AV valve cusps to prevent backflow of blood
(C)	Chordae tendinae	(III)	Protective membrane surrounding the heart
(D)	Bundle of His	(IV)	Conduct electrical impulses to the ventricular musculature

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-IV, B-III, C-I, D-II
(4) A-IV, B-III, C-II, D-I

112. Given below are two statements:

Statement I: The secondary oocyte formed during the first meiotic division retains most of the nutrient-rich cytoplasm of the primary oocyte.

Statement II: The Graafian follicle releases the secondary oocyte during ovulation, which is not surrounded by zona pellucida.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

113. Given below are two statements:

Statement I: Axial skeleton comprises 90 bones distributed along the main axis of the body.

Statement II: The vertebral column is differentiated into cervical (5), thoracic (12), lumbar (7), sacral (1-fused) and coccygeal (1-fused) regions starting from the skull.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

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114. From which of the following ancestral groups did dinosaurs evolve during geological history?
(1) Pelycosaurs (2) Thecodonts
(3) Synapsids (4) Therapsids

115. Identify from the following which belong to same level.
(1) Secondary consumers and herbivores.
(2) Primary consumers and herbivores.
(3) First trophic level and herbivores.
(4) Primary carnivores and second trophic level.

116. The spores in liverworts germinate to form :
(1) free-living gametophytes.
(2) free-living sporophytes.
(3) sporophylls.
(4) gemma.

117. Most crucial step during C_3 cycle is:
(1) decarboxylation and reduction.
(2) reduction and regeneration.
(3) regeneration only.
(4) carboxylation.

118. Read following statements about cockroaches.
A. Each abdominal segment bears a pair of walking legs.
B. The abdomen in both males and females consist of 10 segments.
C. Females bear a pair of short, thread like anal styles which are absent in males.
D. Visceral organs located in the haemocoel are bathed in haemolymph.
E. Hindwings are dark and leathery called tegmina.
Select the **correct** statements:
(1) A and D (2) A and E
(3) B and D (4) C and D

119. Which of the following sequences shows the **correct** development of female gametophyte in flowering plants?
(1) Nucellus $\xrightarrow{meiosis}$ Functional megaspore $\xrightarrow{meiosis}$ Female gametophyte.
(2) Megaspore mother cell $\xrightarrow{\text{meiosis}}$ Functional megaspore $\xrightarrow{\text{mitosis}}$ female gametophyte
(3) Megaspore mother cell $\xrightarrow{meiosis}$ nucellus $\xrightarrow{meiosis}$ megaspore $\xrightarrow{meiosis}$ female gametophyte
(4) Megaspore mother cell $\xrightarrow{meiosis}$ Functional megaspore $\xrightarrow{meiosis}$ female gametophyte

120. Given below are two statements:
Statement I: A sigmoid growth curve is a characteristic of living organism growing in a natural environment.
Statement II: Root elongating at a constant rate shows arithmetic growth.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

121. Match **List-I** with **List-II** w.r.t. arthropods.

List-I		List-II	
(A)	Body covering	(I)	Gills, book gills, book lungs, or tracheal system
(B)	Respiratory organs	(II)	Open type
(C)	Excretory organs	(III)	Chitinous exoskeleton
(D)	Circulatory system	(IV)	Malpighian tubules

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-I, C-III, D-II

122. Choose the **correct** statement with respect to *Mucor*.
(1) Aplanospores are exogenously produced.
(2) Formation of zygospores.
(3) Conidia formation.
(4) Septate mycelium.

123. Branch of taxonomy which is based on cytological information like chromosome number, structure and behaviour is called as:
(1) chemotaxonomy and numerical taxonomy.
(2) numerical taxonomy only.
(3) phylogeny and cytotaxonomy.
(4) cytotaxonomy only.

[Contd...

124. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Tidal Volume (TV)	(I)	2500 mL – 3000 mL
(B)	Inspiratory Reserve Volume (IRV)	(II)	1100 mL – 1200 mL
(C)	Expiratory Reserve Volume (ERV)	(III)	500 mL
(D)	Residual Volume (RV)	(IV)	1000 mL – 1100 mL

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-II, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-I, C-IV, D-II

125. Which of the following is **true** about essential amino acids?

(1) They are synthesized by the human body.
(2) They must be obtained through the diet.
(3) They are not required for protein synthesis.
(4) They are found only in plant-based foods.

126. Which of the following plants produces a drug that does **not** have hallucinogenic effects?

(1) *Erythroxylum coca*
(2) *Atropa belladonna*
(3) *Papaver somniferum*
(4) *Datura*

127. If the age distribution (per cent individuals of a given age or group) is plotted for the population, the resulting structure is called a/an:

(1) population density.
(2) birth rate.
(3) death rate.
(4) age pyramid.

128. O_2 evolved in photosynthesis in green plants comes from water, **not** from CO_2 was proved by:

(1) using a suitable H-acceptor molecule like H_2S.
(2) using radioisotope techniques.
(3) half leaf experiments.
(4) experiments of T. W. Engelmann using *Cladophora* and aerobic bacteria.

129. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Base pairs in each turn of DNA helix (roughly)	(I)	10
(B)	Hydrogen bonds between cytosine and guanine	(II)	2.2
(C)	Length of DNA in humans in metres	(III)	3
(D)	Length of DNA in *E.coli* in mm	(IV)	1.36

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-III, C-II, D-IV

130. Match **List-I** with **List-II**:

List-I		List-II	
(A)	*Aspergillus niger*	(I)	Lactic acid
(B)	*Acetobacter aceti*	(II)	Acetic acid
(C)	*Clostridium butylicum*	(III)	Citric acid
(D)	*Lactobacillus*	(IV)	Butyric acid

Choose the **correct** answer from the options given below.

(1) A-IV, B-II, C-I, D-III
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-III, C-I, D-IV

131. The role of oxytocin in the process of parturition is to:

(1) Inhibit uterine contractions to delay childbirth.
(2) Stimulate the release of colostrum from mammary glands.
(3) Induce stronger uterine contractions, leading to the expulsion of the foetus.
(4) Prevent the expulsion of the placenta after childbirth.

132. A patient suffering from severe dehydration is likely to exhibit which of the following physiological responses?

(1) Decreased release of ADH, leading to increased urine output.
(2) Increased blood volume, leading to inhibition of ADH release.
(3) Suppression of osmoreceptors, leading to decreased blood pressure and GFR.
(4) Activation of osmoreceptors, leading to increased ADH secretion and water reabsorption.

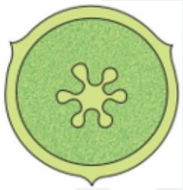
133. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Gastrin	(I)	Acts on exocrine pancreas to secrete water and bicarbonate ions
(B)	Secretin	(II)	Acts on gastric glands to secrete HCl and pepsinogen
(C)	Cholecystokinin (CCK)	(III)	Inhibits gastric secretion and motility
(D)	Gastric Inhibitory Peptide	(IV)	Acts on both pancreas and gall bladder to release enzymes and bile

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-IV, D-II

134. Identify the type of placentation in the given figure with its **correct** example.



(1) Marginal in *Primrose*
(2) Free central in *Dianthus*
(3) Parietal in marigold
(4) Basal in tomato

135. Given below are two statements:

Statement I: The historic Convention on Biological Diversity ('The Earth Summit') was held in Rio de Janeiro.

Statement II: The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

136. Which of the following is the primary reason for the legalization of Medical Termination of Pregnancy (MTP) in India?

(1) To promote population control
(2) To prevent female foeticide
(3) To provide a safe and legal option for terminating unwanted pregnancies
(4) To reduce the burden on healthcare systems

137. Select the **correct** statement.

(1) NAA is a synthetic auxin.
(2) Auxins promote fruit and leaf drop at an early stage.
(3) Auxins prevent the abscission of older mature leaves and fruits.
(4) 2,4-D is a natural auxin.

138. The current species extinction rates are estimated to be _____ faster than in the pre-human times and our activities are responsible for the faster rates.

(1) 10 to 100 times
(2) 100 to 1,000 times
(3) 1,000 to 10,000 times
(4) 10 to 1,000,000 times

139. Isogamous type of sexual reproduction takes place in:

(1) *Volvox* and *Fucus*.
(2) *Fucus* and *Eudorina.*
(3) *Volvox* and *Eudorina.*
(4) *Spirogyra*.

140. The hormone which is produced exclusively during pregnancy and plays a role in maintaining the corpus luteum is:

(1) Progesterone
(2) Human Chorionic Gonadotropin (hCG)
(3) Relaxin
(4) Thyroxine

141. Given below are two statements:

Statement I: The order and sequence of amino acids are defined by the sequence of bases in the mRNA.

Statement II: The amino acids are joined by a phosphodiester bond during translation.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

[Contd...

142. Which of the following statements are **correct** regarding non-chordates?

A. Central nervous system is ventral, solid and double.
B. Gill slits are absent.
C. A post-anal part is present.
D. Notochord absent.
E. Heart is dorsal when present.

(1) A, B, C and D only
(2) A, B, D and E only
(3) B, C, D and E only
(4) A, C, D and E only

143. Read the following and identify the **correct** statements.

I. The effluent from the secondary treatment plant is generally released into natural water bodies.
II. The biogas plant consists of a concrete tank (10-15 feet deep).
III. Statins are produced by the yeast *Monascus purpureus*.
IV. Baculoviruses are pathogens that attack insects and birds.

Choose the *most appropriate* answer from the options given below.

(1) I, II and III (2) Only III
(3) Only I and IV (4) Only II and IV

144. Identify the **correct** statement related to biological classification of wheat.

(1) It belongs to the genus *Mangifera*.
(2) It belongs to the family Convolvulaceae.
(3) It belongs to the order Anacardiaceae.
(4) It belongs to the class Monocotyledonae.

145. The cell wall of kelps is made up of:

(1) cellulose and is covered outside by algin.
(2) cellulose and is covered outside by carrageen.
(3) cellulose and is covered inside by carrageen.
(4) cellulose and is covered inside by algin.

146. Which of the following statements regarding oral contraceptive pills are **correct**?

A. They inhibit ovulation and implantation.
B. They alter the quality of cervical mucus to prevent sperm entry.
C. They must be taken daily for 28 days without any gap.
D. Saheli is a non-steroidal oral contraceptive taken once a week.
E. All oral contraceptive pills contain high doses of estrogen.

(1) A, B and C
(2) C, D and E
(3) A, B and D
(4) B, C and E

147. A scientist cuts two different DNA samples using the same restriction enzyme (*EcoRI*) and mixes them. After adding DNA ligase, what will most likely happen?

(1) The DNA fragments will randomly join with blunt ends.
(2) The complementary sticky ends will first pair via hydrogen bonding, followed by DNA ligase sealing the phosphodiester backbone.
(3) The DNA fragments will not combine due to mismatched sticky ends.
(4) The DNA ligase will create new restriction sites in the DNA.

148. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Gonorrhoea	(I)	Not completely curable STD
(B)	Hepatitis-B	(II)	Transfer of the zygote or early embryos (with upto 8 blastomeres) into the fallopian tube
(C)	ZIFT	(III)	A sperm is directly injected into the ovum
(D)	ICSI	(IV)	Completely curable STD

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-I, D-II

149. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In both lactic acid and alcohol fermentation less than seven per cent of the energy in glucose is released.

Reason R: Net gain of ATP in fermentation is 12 ATP.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

[Contd...

150. Which of the following statements about the hindbrain is **incorrect**?
(1) The hindbrain comprises pons, cerebellum and medulla.
(2) The pons contains fibre tracts that interconnect different regions of the brain.
(3) The cerebellum has a highly convoluted surface to accommodate more neurons.
(4) The medulla oblongata is responsible for controlling voluntary movements.

151. Sickle-cell anaemia is caused by substitution of:
(1) valine by glutamic acid at sixth position of beta globin chain.
(2) glutamic acid by valine at sixth position of beta globin chain.
(3) valine by glutamic acid at seventh position of alpha chain.
(4) glutamic acid by valine at seventh position of alpha chain.

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The R group in amino acids determines whether they are acidic, basic, or neutral.
Reason R: The R group in lysine contains an additional carboxyl group, making it an acidic amino acid.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

153. Kinetin, a modified adenine (purine) was discovered from:
(1) autoclaved herring sperm DNA.
(2) coconut milk.
(3) corn kernel.
(4) fungus.

154. Which of the following is **correct** statement?
(1) von Tschermak coined the term linkage.
(2) Punnett square used to understand a typical monohybrid cross conducted by Mendel.
(3) Gregor Mendel noted that the behaviour of chromosomes was parallel to the behaviour of genes.
(4) Law of segregation was given by Thomas Hunt Morgan.

155. Which of the following statement is **true** regarding taxa?
(1) Taxa always represent the same level of classification.
(2) Taxa can exist at different levels of classification.
(3) Taxa only refer to species-level classification.
(4) Taxa is another term for species.

156. Which structure in frogs receives bile and pancreatic juices through a common bile duct?
(1) Stomach (2) Duodenum
(3) Rectum (4) Cloaca

157. Which of the following statements about the trachea are **correct**?
A. It is supported by incomplete cartilaginous rings.
B. It divides into the right and left primary bronchi at the level of the 5th thoracic vertebra.
C. It is lined with ciliated epithelium to filter and trap dust particles.
D. It is a flexible tube that collapses during expiration to expel air.
E. It comprises actual exchange part of respiratory system.
(1) B and D only (2) C and D only
(3) D and E only (4) A, B and C

158. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Pulmonary circulation	(I)	Supplies oxygenated blood to the heart muscle and returns deoxygenated blood to the heart
(B)	Systemic circulation	(II)	Supplies oxygenated blood to the entire body and returns deoxygenated blood to the heart
(C)	Hepatic portal system	(III)	Carries deoxygenated blood from the heart to the lungs and returns oxygenated blood to the heart
(D)	Coronary system	(IV)	Transports nutrient-rich blood from the digestive organs to the liver for processing

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-IV, B-I, C-II, D-III

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159. The AIDS virus causes deficiency of the immune system in the infected person by:
(1) Destroying B lymphocytes, reducing antibody production.
(2) Infecting and destroying helper T cells, weakening immune response.
(3) Increasing the production of cytotoxic T cells, attacking healthy tissues.
(4) Stimulating excessive antibody production, leading to immune exhaustion

160. Read the following statements about class Cyclostomata.
A. They have a sucking and circular mouth without jaws.
B. Their body is covered with minute placoid scales.
C. They possess a cartilaginous cranium and vertebral column.
D. They are exclusively marine and do not migrate to freshwater.
E. Cyclostomes have a closed circulatory system.
Select the **correct** statements:
(1) A and D (2) A and C only
(3) B and E (4) A, C, and E

161. Which of the following xylem and phloem elements remain living at maturity?
(1) Tracheids and bast fibres.
(2) Vessels and sieve tubes.
(3) Xylem parenchyma and companion cells.
(4) Xylem fibres and phloem parenchyma.

162. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Hyperglycemic hormone	(I)	Parathyroid hormone (PTH)
(B)	Hypercalcemic hormone	(II)	Glucagon
(C)	Gonadotrophic hormone	(III)	Atrial Natriuretic factor (ANF)
(D)	Blood pressure lowering hormone	(IV)	Luteinizing Hormone (LH)

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-II, B-I, C-III, D-IV

163. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Optimum temperature	(I)	Loss of enzyme structure and function
(B)	V_{max}	(II)	Temperature for highest enzyme activity
(C)	Competitive inhibition	(III)	Maximum reaction rate at enzyme saturation
(D)	Denaturation	(IV)	Inhibitor mimics substrate, binds active site

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

164. Species-area relationship is represented on a log scale as:
(1) hyperbola. (2) rectangular hyperbola.
(3) a straight line. (4) inverted curve.

165. Given below are two statements:
Statement I: In many species of fig trees, there is a tight one-to-one relationship with the pollinator species of wasp.
Statement II: A given fig species can be pollinated only by its partner wasp species and no other species.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

166. Read the following statements about male reproductive system.
A. Each testis is approximately 4–5 cm in length and 2–3 cm in width and is covered by a dense protective layer.
B. Leydig cells, located in the interstitial spaces, secrete androgens, the male sex hormones.
C. The male sex accessory ducts include the rete testis, vasa efferentia, epididymis, and vas deferens.
D. The epididymis is located along the anterior surface of the testis and stores mature sperm.
E. The vasa efferentia transport sperm from the epididymis to urethra.
Select the **incorrect** statements:
(1) A and D (2) A, B and C
(3) C and D (4) D and E

167. Which of the following process is represented in the given diagram?



(1) Cutting of DNA at specific locations
(2) Amplification of DNA
(3) Staining of DNA
(4) Collection of isolated DNA by spooling

168. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Chromatids start moving towards opposite poles in telophase.
Reason R: Nuclear envelope develops around the chromosome clusters at each pole in telophase.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

169. The role of Na^+ in the generation of an action potential is:
(1) To maintain the resting membrane potential solely
(2) To initiate depolarization by entering the neuron
(3) To repolarize the membrane by exiting the neuron
(4) To block the action potential conduction

170. In the pBR322 vector, if foreign DNA is inserted into the *Sal* I site within the tetracycline resistance gene, what will be the outcome for the recombinant transformants?
(1) They will be resistant to both ampicillin and tetracycline.
(2) They will be resistant to ampicillin but sensitive to tetracycline.
(3) They will be sensitive to ampicillin but resistant to tetracycline.
(4) They will be sensitive to both ampicillin and tetracycline.

171. Select the **correct** match from the following pairs.
(1) Radial vascular bundle : Dicotyledonous leaf
(2) Polyarch xylem : Dicotyledonous root
(3) Starch sheath : Monocotyledonous root
(4) Well developed pith : Monocotyledonous root

172. Choose the **incorrectly** matched pair.
(1) X-ray → Carcinogens
(2) Benign tumors → Confined to their original location
(3) Malignant tumors → Show metastasis
(4) Oncogenes → Tumor suppressing genes

173. Which of the following statements about sporopollenin is **false**?
(1) Exine of pollen grain is made up of sporopollenin.
(2) Sporopollenin is one of the most resistant organic materials known.
(3) Degradable by acids.
(4) Sporopollenin can withstand high temperatures.

174. Identify the **correct** sequence of electron transport in ETS of thylakoid during non-cyclic photophosphorylation.
(1) PS-II → Cyt B6f → PQ → PC → PS-I
(2) PS-I → PC → Cyt B6f → PQ → PS-II
(3) PS-II → PQ → Cyt B6f → PC → PS-I
(4) PS-I → Cyt B6f → PC → PQ → PS-II

175. The equation of *Verhulst-Pearl* logistic growth is $\frac{dN}{dt}=rN\left[\frac{K-N}{K}\right]$. In this equation, K indicates:
(1) intrinsic rate of natural increase.
(2) biotic potential.
(3) carrying capacity.
(4) population density.

176. According to Paul Ehrlich's "Rivet Popper hypothesis", the rivets on the wings represent:
(1) alien species.
(2) endemic species.
(3) key species.
(4) indigenous species.

177. Which of the following are not Mendelian disorders?
A. Haemophilia
B. Cystic fibrosis
C. Colour blindness
D. Turner's syndrome
E. Down's syndrome
Choose the *most appropriate* answer from the options given below:
(1) A, B and C only
(2) B, C and D only
(3) C, D and E only
(4) D and E only

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178. The number of chromosomes and amount of DNA in maize root tip cell in G_1 phase of cell cycle is 40 and 40 pg respectively. What will be the number of chromosomes and amount of DNA in beginning of G_2 phase?

(1) 20 and 20 pg respectively
(2) 20 and 40 pg respectively
(3) 40 and 80 pg respectively
(4) 40 and 40 pg respectively

179. Scutellum, a structure found in embryo of certain plants is a:

(1) remnant of second cotyledons in monocots.
(2) single cotyledon in monocots.
(3) root primordia in dicots.
(4) single cotyledons in dicots.

180. There is growing public opposition to patents granted on biological resources and genetic materials because:

(1) Patents prevent the use of traditional biological knowledge by local farmers and indigenous communities.
(2) Patents allow free access to genetic resources for all nations.
(3) Genetic modification of organisms does not require ethical consideration.
(4) Patents encourage excessive genetic engineering without legal oversight.

■■■

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